the overall performance by using the 'stacking' and 'voting' techniques. In all these techniques, data is divided into train and test data sets. The model learns on the train data set and is tested on the test data set.

The ensemble methods can be broadly categorised into bagging, boosting, stacking, and voting, as shown in Figure 1.

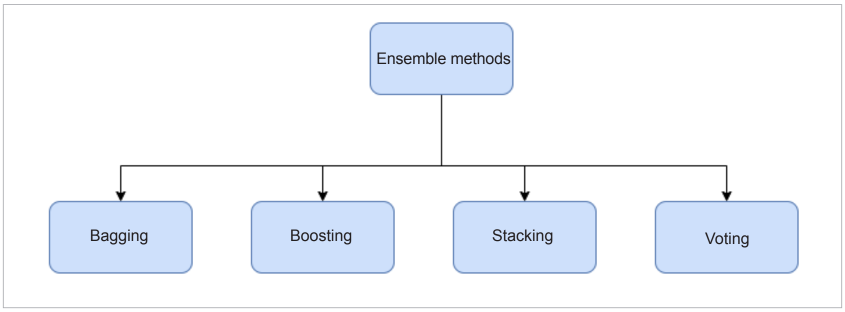

Figure 1: Ensemble methods

Bagging

Bagging is a two phase 'parallel learning' technique, as shown in Figure 2.

a. The first step is bootstrapping. This is a statistical technique, which involves generating sub-samples from the original data sets (with replacement). This data is fed to the different models for their individual prediction.

b. The next step is aggregations, which takes predicted outputs from each weak model and combines them into strong final predictions, mostly using a voting process.

Since bagging aggregates models, not all the features in the data are present in each bootstrap sample; instead, only a subset of a given size is used. This technique reduces the problem of over-fitting by giving low variance, since the model is trained on different samples of the same data sets and the result is an aggregation of the individual model's output.

A very popular example of a bagging technique is random forest.

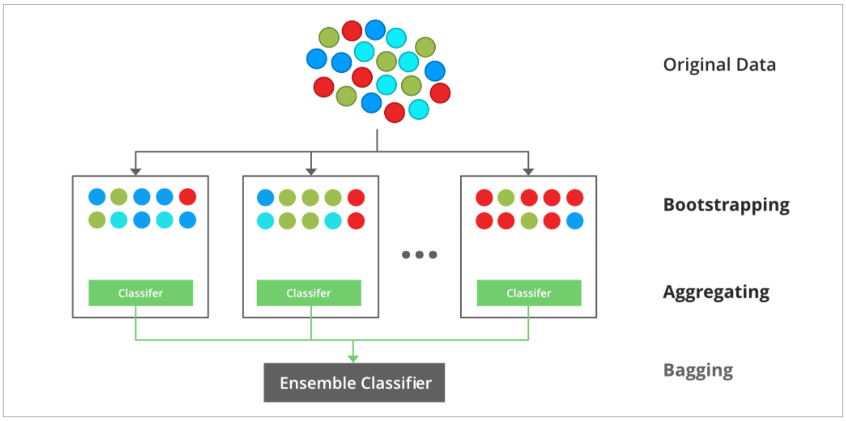

Figure 2: Bagging

It uses decision tree as the inner model. Decision trees generally have low bias but high variance. Bagging reduces variance and keeps the bias low. This ensemble method greatly enhances the power of the decision tree and produces a better model.

Boosting

In this technique, several instances of the same base model are trained sequentially.

The following steps are involved.

1. The first step is to initialise the data set and assign equal weights to each of the data points.
2. The second step is to train it with the model that best classifies data with accuracy. This is provided as an input to the model and the wrongly classified data points are identified.
3. In the next step, the weights of the wrongly classified data points are

Table 1: Bagging vs boosting

Bagging	Boosting
Focus on decreasing variance	Focus on decreasing bias
Each inner model receives equal weight	The inner models are weighted according to their performance
Each model is built independently	New models are influenced by the performance of previously built models
Different training data subsets are selected using row sampling with replacement	Every new subset contains the elements that were misclassified by previous models
If the classifier is unstable (high variance), then bagging is applied	If the classifier is stable and simple (high bias), then boosting is applied
In this base, classifiers are trained in parallel	In this base, classifiers are trained sequentially